```
%run "../includes/configuration"
```

Agrregate functions demo

Build in Aggregate functions

```
race_results_df = spark.read.parquet(f"{presentation_folder_path}/race_results")
```

```
display(race_results_df)
```

Table

	race_year	race_name	race_date	circuit_location	driver_name	driver_nationality
69	2009	Bahrain Grand Prix	2009-04-26T12:00:00.000+00:...	Sakhir	Heikki Kovalainen	Finnish
70	2009	Bahrain Grand Prix	2009-04-26T12:00:00.000+00:...	Sakhir	Mark Webber	Australian
71	2009	Bahrain Grand Prix	2009-04-26T12:00:00.000+00:...	Sakhir	Nelson Piquet Jr.	Brazilian
72	2009	Bahrain Grand Prix	2009-04-26T12:00:00.000+00:...	Sakhir	Nico Rosberg	German
73	2009	Bahrain Grand Prix	2009-04-26T12:00:00.000+00:...	Sakhir	Fernando Alonso	Spanish
74	2009	Bahrain Grand Prix	2009-04-26T12:00:00.000+00:...	Sakhir	Timo Glock	German
75	2009	Bahrain Grand Prix	2009-04-26T12:00:00.000+00:...	Sakhir	Kimi Räikkönen	Finnish
76	2009	Bahrain Grand Prix	2009-04-26T12:00:00.000+00:...	Sakhir	Rubens Barrichello	Brazilian
77	2009	Bahrain Grand Prix	2009-04-26T12:00:00.000+00:...	Sakhir	Lewis Hamilton	British
78	2009	Bahrain Grand Prix	2009-04-26T12:00:00.000+00:...	Sakhir	Jarno Trulli	Italian
79	2009	Bahrain Grand Prix	2009-04-26T12:00:00.000+00:...	Sakhir	Sebastian Vettel	German
80	2009	Bahrain Grand Prix	2009-04-26T12:00:00.000+00:...	Sakhir	Jenson Button	British
81	2009	Spanish Grand Prix	2009-05-10T12:00:00.000+00:...	Montmeló	Adrian Sutil	German
82	2009	Spanish Grand Prix	2009-05-10T12:00:00.000+00:...	Montmeló	Sébastien Bourdais	French
83	2009	Spanish Grand Prix	2009-05-10T12:00:00.000+00:...	Montmeló	Sébastien Buemi	Swiss
84	2009	Spanish Grand Prix	2009-05-10T12:00:00.000+00:...	Montmeló	Jarno Trulli	Italian

10,000+ rows | Truncated data due to row limit

```
demo_df = race_results_df.filter("race_year == 2020")
```

```
from pyspark.sql.functions import count, countDistinct, sum
```

```
demo_df.select(count("*")).show()
```

```
+--------+
|count(1)|
+--------+
|     340|
+--------+
```

```
demo_df.select(count("race_name")).show()
```

```
+----------------+
|count(race_name)|
+----------------+
|             340|
+----------------+
```

```
demo_df.select(countDistinct("race_name")).show()
```

```
+-------------------------+
|count(DISTINCT race_name)|
+-------------------------+
|                       17|
+-------------------------+
```

```
demo_df.select(sum("points")).show()
```

```
+-----------+
|sum(points)|
+-----------+
|     1734.0|
+-----------+
```

sum of points scored by driver Lewis Hamilton

```
demo_df.filter("driver_name = 'Lewis Hamilton'").select(sum("points")).show()
```

```
+-----------+
|sum(points)|
+-----------+
|      347.0|
+-----------+
```

we can use multiple functions

```
demo_df.filter("driver_name = 'Lewis Hamilton'").select(sum("points").alias("total_points"), countDistinct("race_name").alias("numb
```

```
+------------+---------------+
|total_points|number_of_races|
+------------+---------------+
|       347.0|             16|
+------------+---------------+
```

Group By

```
demo_df \
    .groupBy("driver_name") \
    .sum("points")  \
    .show()
```

```
|   Kevin Magnussen|        1.0|
|      Sergio Pérez|      125.0|
|      Carlos Sainz|      105.0|
|    Kimi Räikkönen|        4.0|
|   Romain Grosjean|        2.0|
|   Charles Leclerc|       98.0|
|   Alexander Albon|      105.0|
|      Lance Stroll|       75.0|
|      Pierre Gasly|       75.0|
|    Lewis Hamilton|      347.0|
|   Nico Hülkenberg|       10.0|
|  Daniel Ricciardo|      119.0|
|   Valtteri Bottas|      223.0|
|Antonio Giovinazzi|        4.0|
|      Lando Norris|       97.0|
|    Max Verstappen|      214.0|
|    George Russell|        3.0|
|  Sebastian Vettel|       33.0|
+------------------+-----------+
only showing top 20 rows
```

Aggregations under Group By

```
demo_df \
    .groupBy("driver_name") \
    .agg(sum("points").alias("total_points"), countDistinct("race_name").alias("number_of_races")) \
    .show()
```

```
|   Kevin Magnussen|         1.0|              17|
|      Sergio Pérez|       125.0|              15|
|      Carlos Sainz|       105.0|              17|
|    Kimi Räikkönen|         4.0|              17|
|   Romain Grosjean|         2.0|              15|
|   Charles Leclerc|        98.0|              17|
|   Alexander Albon|       105.0|              17|
|      Lance Stroll|        75.0|              16|
|      Pierre Gasly|        75.0|              17|
|    Lewis Hamilton|       347.0|              16|
|   Nico Hülkenberg|        10.0|               3|
|  Daniel Ricciardo|       119.0|              17|
|   Valtteri Bottas|       223.0|              17|
|Antonio Giovinazzi|         4.0|              17|
|      Lando Norris|        97.0|              17|
|    Max Verstappen|       214.0|              17|
|    George Russell|         3.0|              17|
|  Sebastian Vettel|        33.0|              17|
+------------------+------------+---------------+
only showing top 20 rows
```

Windows Functions

```
demo_df = race_results_df.filter("race_year in (2019,2020)")
```

```
display(demo_df)
```

Table

	race_year	race_name	race_date	circuit_location	driver_name	driver_nationality
1	2019	Australian Grand Prix	2019-03-17T05:10:00.000+00:...	Melbourne	Carlos Sainz	Spanish
2	2019	Australian Grand Prix	2019-03-17T05:10:00.000+00:...	Melbourne	Daniel Ricciardo	Australian
3	2019	Australian Grand Prix	2019-03-17T05:10:00.000+00:...	Melbourne	Romain Grosjean	French
4	2019	Australian Grand Prix	2019-03-17T05:10:00.000+00:...	Melbourne	Robert Kubica	Polish
5	2019	Australian Grand Prix	2019-03-17T05:10:00.000+00:...	Melbourne	George Russell	British
6	2019	Australian Grand Prix	2019-03-17T05:10:00.000+00:...	Melbourne	Antonio Giovinazzi	Italian
7	2019	Australian Grand Prix	2019-03-17T05:10:00.000+00:...	Melbourne	Alexander Albon	Thai
8	2019	Australian Grand Prix	2019-03-17T05:10:00.000+00:...	Melbourne	Sergio Pérez	Mexican
9	2019	Australian Grand Prix	2019-03-17T05:10:00.000+00:...	Melbourne	Lando Norris	British
10	2019	Australian Grand Prix	2019-03-17T05:10:00.000+00:...	Melbourne	Pierre Gasly	French
11	2019	Australian Grand Prix	2019-03-17T05:10:00.000+00:...	Melbourne	Daniil Kvyat	Russian
12	2019	Australian Grand Prix	2019-03-17T05:10:00.000+00:...	Melbourne	Lance Stroll	Canadian
13	2019	Australian Grand Prix	2019-03-17T05:10:00.000+00:...	Melbourne	Kimi Räikkönen	Finnish
14	2019	Australian Grand Prix	2019-03-17T05:10:00.000+00:...	Melbourne	Nico Hülkenberg	German

760 rows

```
demo_grouped_df = demo_df \
    .groupBy("race_year", "driver_name") \
    .agg(sum("points").alias("total_points"), countDistinct("race_name").alias("number_of_races")) \
    .show()
```

```
|     2020|   Kevin Magnussen|         1.0|              17|
|     2020|Antonio Giovinazzi|         4.0|              17|
|     2020|   Nico Hülkenberg|        10.0|               3|
|     2020|   Romain Grosjean|         2.0|              15|
|     2019|     Robert Kubica|         1.0|              21|
|     2020|   Charles Leclerc|        98.0|              17|
|     2019|      Lance Stroll|        21.0|              21|
|     2020|      Esteban Ocon|        62.0|              17|
|     2020| Pietro Fittipaldi|         0.0|               2|
|     2019|      Lando Norris|        49.0|              21|
|     2019|  Sebastian Vettel|       240.0|              21|
|     2019|    George Russell|         0.0|              21|
|     2019|      Carlos Sainz|        96.0|              21|
|     2020|  Sebastian Vettel|        33.0|              17|
|     2020|  Daniel Ricciardo|       119.0|              17|
|     2020|    Kimi Räikkönen|         4.0|              17|
|     2019|    Max Verstappen|       278.0|              21|
|     2019|   Valtteri Bottas|       326.0|              21|
+---------+------------------+------------+----------------+
only showing top 20 rows
```

```
demo_grouped_df = demo_df \
    .groupBy("race_year", "driver_name") \
    .agg(sum("points").alias("total_points"), countDistinct("race_name").alias("number_of_races"))
```

```
display(demo_grouped_df)
```

Table

	race_year	driver_name	total_points	number_of_races
1	2020	Daniil Kvyat	32	17
2	2019	Kevin Magnussen	20	21
3	2020	Kevin Magnussen	1	17
4	2020	Antonio Giovinazzi	4	17
5	2020	Nico Hülkenberg	10	3
6	2020	Romain Grosjean	2	15
7	2019	Robert Kubica	1	21
8	2020	Charles Leclerc	98	17
9	2019	Lance Stroll	21	21
10	2020	Esteban Ocon	62	17
11	2020	Pietro Fittipaldi	0	2
12	2019	Lando Norris	49	21
13	2019	Sebastian Vettel	240	21
14	2019	George Russell	0	21
15	2019	Carlos Sainz	96	21

43 rows

Rank

```
from pyspark.sql.window import Window
from pyspark.sql.functions import desc, rank, dense_rank

driverRankSpec = Window.partitionBy("race_year").orderBy(desc("total_points"))
demo_grouped_df.withColumn("rank", rank().over(driverRankSpec)).show(100)
```

```
|     2020|  Daniel Ricciardo|       119.0|             17|   5|
|     2020|      Carlos Sainz|       105.0|             17|   6|
|     2020|   Alexander Albon|       105.0|             17|   6|
|     2020|   Charles Leclerc|        98.0|             17|   8|
|     2020|      Lando Norris|        97.0|             17|   9|
|     2020|      Pierre Gasly|        75.0|             17|  10|
|     2020|      Lance Stroll|        75.0|             16|  10|
|     2020|      Esteban Ocon|        62.0|             17|  12|
|     2020|  Sebastian Vettel|        33.0|             17|  13|
|     2020|      Daniil Kvyat|        32.0|             17|  14|
|     2020|   Nico Hülkenberg|        10.0|              3|  15|
|     2020|Antonio Giovinazzi|         4.0|             17|  16|
|     2020|    Kimi Räikkönen|         4.0|             17|  16|
|     2020|    George Russell|         3.0|             17|  18|
|     2020|   Romain Grosjean|         2.0|             15|  19|
|     2020|   Kevin Magnussen|         1.0|             17|  20|
|     2020| Pietro Fittipaldi|         0.0|              2|  21|
|     2020|       Jack Aitken|         0.0|              1|  21|
|     2020|   Nicholas Latifi|         0.0|             17|  21|
+---------+------------------+------------+---------------+----+
```

Dense Rank

```
from pyspark.sql.window import Window
from pyspark.sql.functions import desc, rank, dense_rank

driverRankSpec = Window.partitionBy("race_year").orderBy(desc("total_points"))
demo_grouped_df.withColumn("rank", dense_rank().over(driverRankSpec)).show(100)
```

```
|     2020|  Daniel Ricciardo|       119.0|             17|   5|
|     2020|      Carlos Sainz|       105.0|             17|   6|
|     2020|   Alexander Albon|       105.0|             17|   6|
|     2020|   Charles Leclerc|        98.0|             17|   7|
|     2020|      Lando Norris|        97.0|             17|   8|
|     2020|      Pierre Gasly|        75.0|             17|   9|
|     2020|      Lance Stroll|        75.0|             16|   9|
|     2020|      Esteban Ocon|        62.0|             17|  10|
|     2020|  Sebastian Vettel|        33.0|             17|  11|
|     2020|      Daniil Kvyat|        32.0|             17|  12|
|     2020|   Nico Hülkenberg|        10.0|              3|  13|
|     2020|Antonio Giovinazzi|         4.0|             17|  14|
|     2020|    Kimi Räikkönen|         4.0|             17|  14|
|     2020|    George Russell|         3.0|             17|  15|
|     2020|   Romain Grosjean|         2.0|             15|  16|
|     2020|   Kevin Magnussen|         1.0|             17|  17|
|     2020| Pietro Fittipaldi|         0.0|              2|  18|
|     2020|       Jack Aitken|         0.0|              1|  18|
|     2020|   Nicholas Latifi|         0.0|             17|  18|
+---------+------------------+------------+---------------+----+
```